

OMNI — an object-based model of the access network infrastructure

R M Redstall, J R Reid, R Mardani and J Mellis

An object-based software tool is described for the calculation of network performance in the access infrastructure network. Such a tool is representative of a number of new functions and features which have potential for addition to BT's future software-based planning systems. A prototype software model has been created to represent the network components, and has allowed a number of issues to be assessed, such as the design of the data structure, the links to external databases, the calculation and display of reliability data, alternative network representation, and the role of geographical and schematic maps in network design.

1. Introduction

The need to introduce automation into the process of planning the physical infrastructure in the access network is driven by motives which include the wish to improve planning response times, to carry out more complex planning, to ensure consistency, to forecast customer capacity requirements, to establish optimum routeing and records information, and to increase efficiency. Until now, initiatives to achieve this have concentrated on using the traditional planning methods of capacity maximisation and cost minimisation. Further functions and features, such as the assessment of network reliability, whole-life cost and transmission performance, can also be built into the planning tools. The general benefit will be a reduction in operating costs, and improved network designs.

These extra functions and features have been explored in this work by constructing an object-based software model of the access network infrastructure (OMNI), along with the necessary databases. This has established how the data interacts with such a model through the graphical display of the objects in the network, and how to compose drawings of the network of graphical objects. In this instance, these drawings are the basis for calculations of network reliability, because the graphical objects in the display have associated attributes which allow the calculations to be made, by using a set of stored formulae (methods). Other performance factors (transmission parameters, costs, etc) could be treated similarly, in a comprehensive performance model.

The feasibility of the object-based approach was explored with the aid of the OMNI prototype.

2. Universal performance issues

The initiative for this work has come from the desire to make available a wider range of performance criteria for network planners. The historical position has been that planners have made good use of capital cost information, although whole-life costs are usually not considered. Some of the earlier methods used to calculate access fibre network performance have been augmented with the recent introduction of the fibre PON loss model [1]. Here the motivation has been to seek new tools to improve the efficiency of the planning process further by showing how network reliability, for example, could be added to the process of planning the physical infrastructure. This has demonstrated what can be achieved, with the lessons learned having a broader relevance, that can be applied to wider performance measures.

This paper describes how two tasks were drawn together. The first was to convert the calculations of the reliability of the physical infrastructure from their present spreadsheet form into a graphical, object-based form. This needs to have sufficient flexibility to calculate the reliability result for an infrastructure network as it is being detailed on a computer screen.

The second task concerned how to display the network and the results of the calculations. The necessary series of graphical user interfaces were created, including the display of the network, and the results of any calculations shown in a suitable form for a non-expert user.

The intention is to apply the approach to both metallic and fibre networks, but this paper concentrates on fibre systems, and the only performance measure dealt with so far is network reliability. A typical fibre network is described below.

3. The infrastructure network

The infrastructure used as a reference in this assessment work is based on a typical passive optical network (PON) [2, 3] which is shown schematically in Fig 1, with the cable lengths indicated. The PON architecture shown in Fig 1 utilises a separately routed ring in the primary network, which is interspersed with a series of primary nodes, where each node feeds a separate PON. This first node (node 1 in Fig 1) in the circuit will often be a 4-way splitter. From each leg of those splitters the unduplicated secondary cable joins to the secondary node (node 2 in Fig 1) where there is an 8-way splitter, giving a total split of 32. Finally, the drop cable reaches the customer through a lead-in connection point.

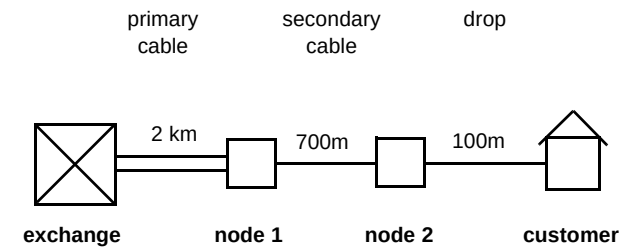


Fig 1 Reference fibre network with a duplicated primary cable.

Relevant data has to be attached to each of these network components, such as their reliability, transmission loss, and how many connections can be made to them, along with the nature of their relationship to each other.

3.1 Infrastructure objects

A simplified generic palette of icons of the objects required was created. Objects may be picked from the palette and then placed on the screen to build up the network. Specific examples of the infrastructure objects can be augmented from larger lists of objects drawn from external databases — a limited set of thirteen was found to be sufficient for the generic objects in the database (see Table 1). Behind these lie larger databases containing all the possible examples of each type of object.

Nodes and links require a different number of connections. Links are objects with two connections — one at each end. Nodes may have more than two connections.

Table 1 Generic infrastructure fibre objects

Links	In-exchange fibre cable
	Primary 96-way fibre cable
	Secondary blown fibre 8-way fibre drop
	Fibre drop
Nodes	SOSA 8-tray
	SOSA 4-tray
	SASA WDM exchange-based
	SASA 2:2 splitter
	SASA 2:4 splitter
	SASA 2:8 splitter
	SASA 2:16 splitter
	SASA 2:32 splitter
	CLI — customer lead-in

3.1.1 Hierarchy of the infrastructure objects

A hierarchy can be defined for the objects in the physical infrastructure network. Cables are contained within the ducts; each cable then contains the fibres or copper pairs, and so on. Ultimately, the full utilisation of object oriented software will allow best use of the hierarchy that exists between the objects. However, while the prototype has employed a compromise by working at only one level in the hierarchy, the data structure is designed to describe the objects more rigorously. An important feature of the objects in the different hierarchical classes is that they can inherit each other's properties when required. This is an important benefit of using object oriented software.

The hierarchy of the infrastructure objects is shown in Fig 2.

A key issue is that the reliability of a circuit connection is not necessarily affected by damage to higher levels in the hierarchy. For instance, a broken duct bore may still leave the circuit intact. The hierarchical approach to representing these various network infrastructure elements offers valuable flexibility when representing issues such as this.

3.1.2 Features of the objects

The inputs to the software model were made through a series of custom-made layouts for the graphical user interface. Certain instances of each object in the database can be changed, such as its length. The objects are described with a unique code number, supplier name, insertion loss (dB), reliability failure rate (FITs), cost and connectivity information as defined by the graphical layout.

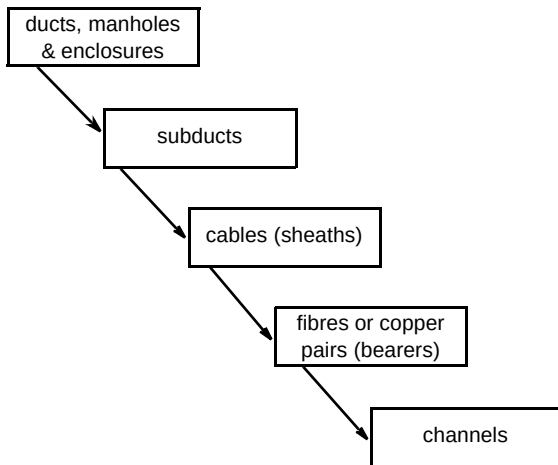


Fig 2 Infrastructure hierarchy.

Much of the data has its source in a relational form — statistics, lengths, losses, etc (from sources such as Excel spreadsheets and company business databases). However, to use the chosen data in this model, it is first drawn into an intermediate database, which is the point at which the attributes of each object can be defined. This is a hybrid relational/object-oriented approach, which was appropriate for this application.

3.2 Design of the software

The value of object orientation (OO) is that the objects and the algorithms behind the calculations made can be defined easily. They may then be re-applied to objects in different networks without any further modification. The alternative and traditional approach for programming requires that, for each new network and its objects, the data

and methods have to be re-constructed each time. Thus OO offers flexibility, reusability and a saving of development time. Care has to be taken to define the nature of the objects sufficiently clearly and robustly so that the objects can be widely applied to the chosen networks.

The three principles embodied in the OO approach are encapsulation, inheritance and polymorphism. In this method, data describing an object is said to be within an object. The principle of encapsulation defines how the data within the object can be related to the methods describing how the data is used in calculation. An object has certain properties, functions and data, and inheritance describes how these are passed to other levels in the hierarchy. Polymorphism is the ability to use the same function across the hierarchy to achieve different behaviour, with a different result in each case. These properties make the object oriented approach attractive here, and it has been applied to the physical infrastructure.

3.2.1 The modelling process

The software [4] was designed to offer a friendly graphical user interface which represented the network by drawing the objects — the nodes and links — on a screen.

Physical infrastructure performance calculations are made with data that is attached to each object, and using algorithms that have been built into the software. The functional blocks in the program are indicated in Fig 3.

The model functions by taking data about the network objects as they are chosen, and adding data from other local databases. This is converted into an object oriented database for use in the model, where behaviour, such as the connectivity information, is added. The user can add new

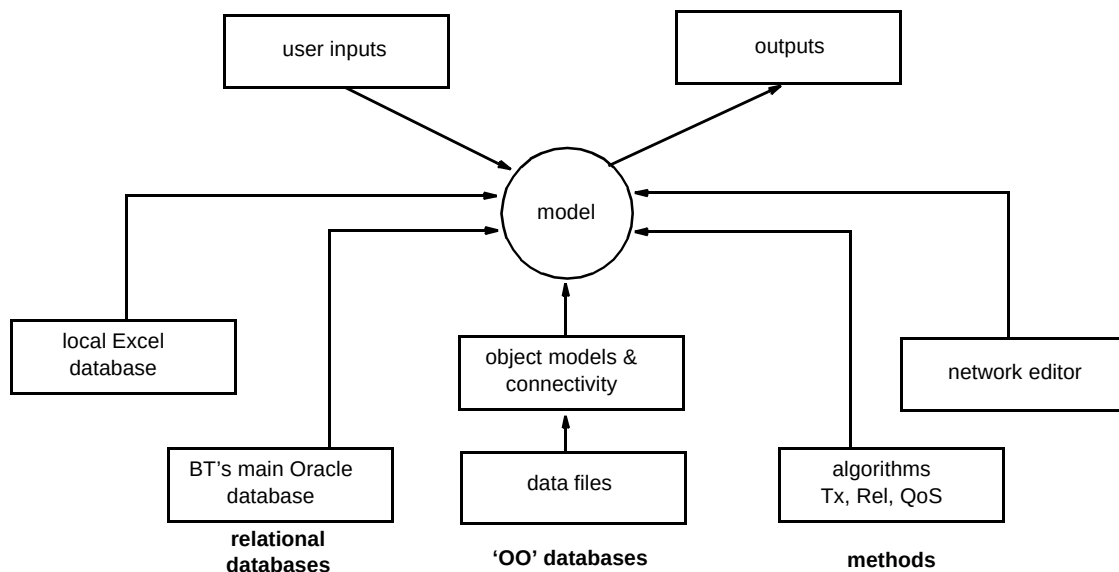


Fig 3 Program functional blocks.

connectivity information, as the network editor creates the graphical representation of the network, and will then be able to use the predetermined algorithms as required. The steps to create the network are described below.

3.2.2 Representing the network

The software provides a flexible graphical user interface. There are three types of interface screen required:

- the graphical representation of the components themselves,
- the actual drawings of network information, such as ducts or circuits,
- the background information, such as Ordnance Survey maps.

The infrastructure objects are picked from the palette and placed on the screen, and the network is built up from the exchange, using cables to join the nodes, out to the customer connection.

The transmission bearer which is to be displayed on the screen, depicted in Fig 4, is composed of links and nodes. In this example, a PON deployment is illustrated, where the

primary optical cable forms a ring from the exchange (N1 nodes), and feeds nine primary nodes (N1), each of which lead to secondary nodes (N2) via unduplicated fibre, and thence to the individual customers. The primary cable uses a 96-way fibre, and the nodes contain optical splitters which are selected from 2-, 4-, 8-, or 16-way splitters to give a 32-way split overall.

The example in Fig 4 also displays an Ordnance Survey map behind the infrastructure duct and cable network. When these duct networks are to scale, duct lengths may then be determined automatically.

3.2.3 Selecting a network portion for calculation

Once the network has been created within the network editor using the chosen objects, an important function will be the method by which a circuit or path to a chosen customer in the network will be selected, before the performance can be calculated.

The result will differ depending on four factors, which are:

- network configuration,
- customer location,

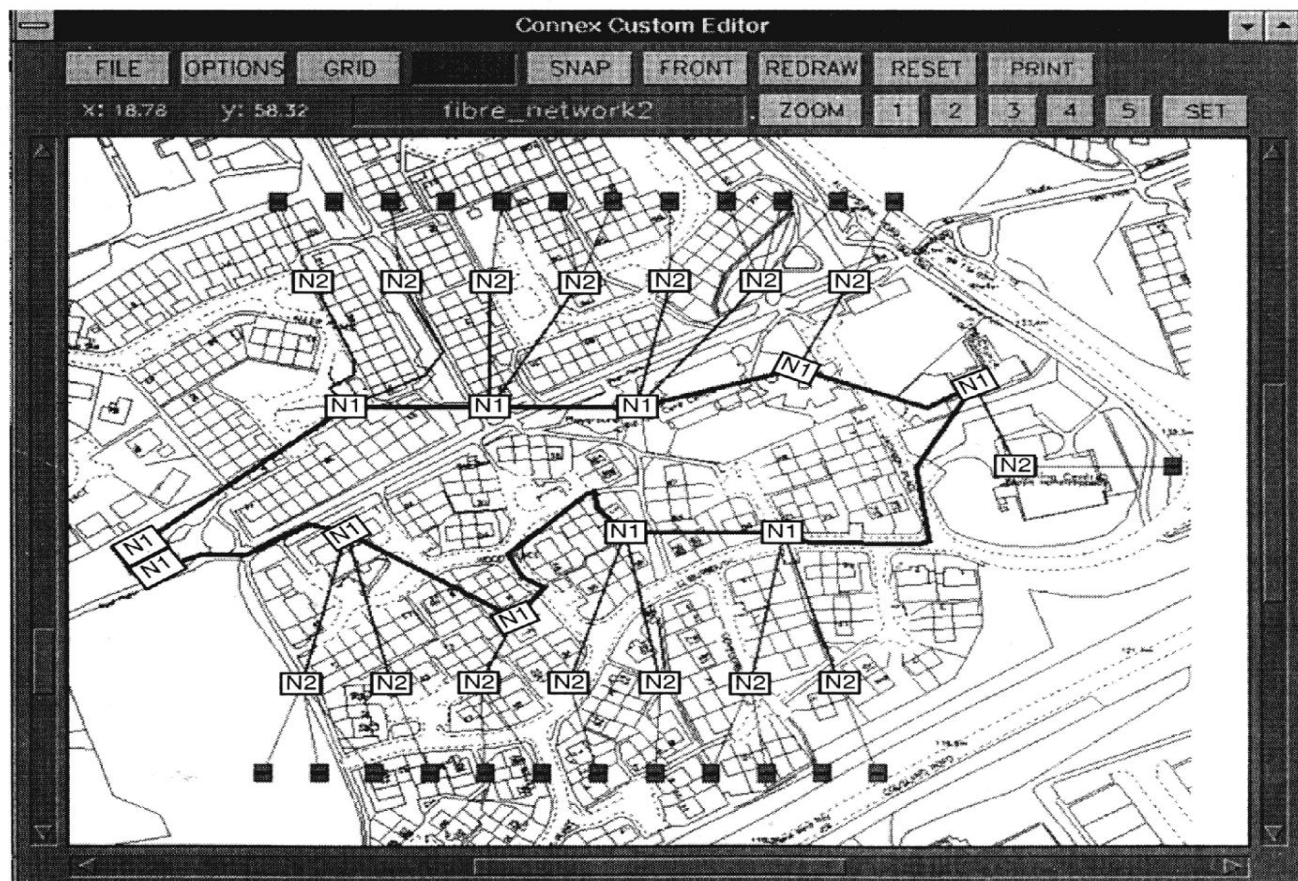


Fig 4 Part of a typical network graphical user interface.

- bearer routing,
- distance.

The point around the primary ring to which the customer's PON is connected will influence the result. The shorter the distance of the main link to the exchange, the better the projected reliability. However, the main and stand-by links will be switched back and forth periodically because of external factors, so that both values will need to be calculated, and the average value quoted.

3.2.4 Editing the assumptions

A number of pre-set assumptions will be embedded in a reference section of any model of component and network reliability. These will include component failure rate, cable damage intervals, duplication factors and average operating temperatures.

Because the primary cable route offers a level of resilience, the benefit of this is calculated with a particular reliability algorithm. Generally, the main and stand-by parts of this resilient link have different lengths, and an important feature of any model will be to show the benefits of resilience in different parts of a network. Also important are the assumptions about the effect of environmental factors, for instance, whether the cables and nodes will be underground or overhead, and whether other parts of the infrastructure will be located in the exchange or in outside enclosures.

4. Functions of the reliability calculations

The network drawing in Fig 1 is the basis for listing each of the components or objects in the network. The individual failure rates are allocated for each object. In the case of a cable, for instance, the failure rate comes from knowing how often the cable is damaged, dug up by third parties or fails intrinsically. When the failure rates are multiplied by the restore times, the result is the 'downtime' for the object. The physical infrastructure reliability is obtained by adding the downtimes to give the overall downtime for the network (this is usually expressed in minutes per year).

These reliability calculations are able to give additional quality of service (QoS) measures, from the details that lie behind this single downtime figure. One of these is to derive an estimate of the effort needed to keep the whole network operational. From a knowledge of the total network reliability, coupled with the associated repair times for each network element, the resulting 'maintenance burden' of the network can be calculated. In turn, that can lead to an estimated current account expenditure to maintain the chosen network on an annual basis.

A further point is the issue of a 'repair-induced-break'. Here, an allowance is made for the service interruption that a customer with a working circuit may experience when the repair of a piece of common equipment occurs. An example of that is when one leg of a splitter breaks, and all the working circuits common to that splitter suffer a break of service during its subsequent replacement.

Another important measure of network QoS is 'availability'. This is currently used for different products to specify service targets, although it is an oversimplification to represent network QoS with such a single number.

For tactical reasons, another requirement of the modelling tool is to be able to explore 'what-if' assessments on the effect of changing components or objects in the network, such as varying cable lengths, removing duplication, etc.

In the prototype, the calculations are made on the basis of the circuit connections to customers. This work has shown that the physical infrastructure reliability calculations can be made to function at the bearer level only. Any additional reliability contributions from the other levels in the hierarchy have been taken into account by adding their influence specifically when the code is written. For example, the unreliability originating from third-party damage to ducts and cables is added into the algorithm for the fibre reliability, rather than it being drawn in automatically as an attribute of the fibre link. In the same way, the different environments of the objects have been allowed for directly by adjusting the failure rates held in the database. The more general approach, to utilise the full hierarchy of the objects, will be investigated in future work.

5. Features of the displayed results

The features which form the displayed output can be designed to display or conceal many outputs. Future work will need not only to assess how these outputs could best be summarised, but also to explore the usefulness of the calculations in network planning.

Network reliability can be presented in five alternative ways:

- total network failure rate (expressed either as a rate, or in FITs — one FIT is a failure rate of one in 10^9 hours),
- failure rate for the circuit (also either as a failure rate, or in FITs),
- downtime expressed in minutes,
- availability,
- unavailability.

These five reliability numbers are all related, coming originally from the predicted failure rate for that component, and from the additional repair-induced breaks caused when changing shared components.

A further possibility is the calculation of the overall maintenance burden for the PON network. This can be displayed in two ways, either as an overall failure rate — a total number of FITs (e.g. 60 000 FITs), or as an equivalent mean time before failure (e.g. MTBF 1.9 years). The maintenance burden for the PON in manhour activity per year can be converted into the annual expenditure to maintain that network (with the knowledge of the hourly labour rates and stores costs).

6. Conclusions

New methods of modelling performance have been tested to enable them to be better incorporated in future software tools, and to assist the planning process for the access infrastructure. The network objects used to represent the network have been defined, and the reliability algorithms specified. The hierarchy of the objects has also been defined, and a basic library of external plant objects established.

A prototype software performance model of the physical infrastructure has been successfully used to assess the feasibility of specifying the functions and features of the infrastructure network reliability and to demonstrate the object-based approach.

The performance model will need further assessment, comparison with alternative models, and general refinement and development to explore the use of the object library in network planning.

Acknowledgements

The authors are grateful for the assistance of their colleagues, Richard Colver and Richard Walling.

References

- 1 Bell P et al: 'Optical power budget calculating tool for fibre in the access network', *BT Technol J*, **14**, No 2, pp 116—120 (April 1996).
- 2 Redstall R M: 'PON system reliability', IEE paper (February 1994).
- 3 Mellis J, Redstall M, Shaw R N and Rabone N: 'Performance and reliability requirements for passive components in local optical networks', *EFOC* (1994).
- 4 Tenet Systems, Horsham, UK.



Martin Redstall joined BT Laboratories in 1973, analysing epitaxial III-V semiconductors, after graduating with a BA in Physics (and later an MSc in solid state physics) from Essex University. Subsequently he worked on the reliability of semiconductor plastic packaging, IR semiconductor laser chips and packaging, and passive optical splitters.

He was a member of a European RACE project, concerned with the reliability of optical splitters, and the overall network reliability of fibre systems.

He presently works on advanced methods and tools for planning fibre access networks in the external plant and pilots unit.



John Reid graduated in 1986 with an honours degree in physics from Edinburgh University. Since joining BT in 1986, he has worked in a number of areas, including blown fibre development, optical connector technology and in-building wiring.

He studied part time for an MSc in telecommunications at Essex University which he completed in 1992, and subsequently joined the IEE as a corporate member.

He is currently with the external plant and pilots unit working in the area of access network infrastructure planning and performance, and leads a project investigating opportunities for planning automation.



Reza Mardani graduated in 1986 with a BSc(Hons) in electrical engineering and a PhD in high-resolution array signal processing in 1990 from Southampton University.

He worked in the field of control and computing at National Power PLC. From 1993—95 he was with BT Laboratories and developed object oriented/relational models for copper and fibre networks. His research interests include signal processing, linear and nonlinear system identification, and control and mathematical/software modelling, and he is an associate member of the IEE.



John Mellis leads the external plant and pilots unit at BT Laboratories, which specialises in the performance, planning, and pilot trial of external network infrastructure components and systems. He joined BT in 1986 to engineer network demonstrations of coherent optical transmission, and from 1989—91 was with BT&D (now HP Optoelectronics), commercialising optical amplifier and tunable laser products. From 1983—86 he was at STL, Harlow (BNR Europe), developing electro-optic/acoustic-optic signal processors.

He received a doctorate in 1983 from the University of St Andrews, for the mathematical modelling of CO₂ laser kinetics, and graduated in applied physics (with operational research) from the University of Strathclyde in 1978.

He is chairman of the ETSI expert group on passive optical components, sits on the technical advisory committee of the European Conference on Networks and Optical Communications, and is a Member of the IEEE.